IC-PBL Project of Machine Learning 2025-Fall

Topic: Building an explainable, efficient classifier for human exercise motions

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Supported by: Hexar Humancare
Industry Mentor: Dr. Young Hoon Ji

1. Description
   Rehabilitation exercises consist of a few sets of human motions, which were designed to recover the target muscle or linkages of human body. However, due to various reasons including injuries, aging, or after-surgery recovery, many data gathered in practice are difficult to recognize and classify. In this study, within the scope of this course, we simplified the problem as building an "explainable" classifier for such recorded trajectory data to classify "which" of the motions the users (patients) made.

2. Dataset
   You'll be given two sets of data, each of which contains a small number of trajectory (series) data which describes different human exercise motions. The first set (distributed on 11th week) consists of data without much noise (relatively good quality), and the second set (will be distributed on 13th week) consists of noisy, imperfect data. For detailed explanations, please refer to the LMS announcement by our TA, Minho Chung.
   Given that the dataset has a small number of "series." Thus, you are asked to *use* the data wisely. You may augment, enhance, or do some engineering by yourself if needed.

3. Deliverables
   0) Each team must prepare three mentoring reports (according to We-Meet Program of the university), announced on LMS.
   1) Final products (Due on Dec 16, BEFORE class) include:
      A. The code of the AI model or algorithm the team built
      B. Final report, which includes:
         i. Objectives

ii. Approaches and Idea, and theoretical backgrounds regarding them
iii. Explanations on the team's implementation and how" it works
iv. Discussions on edge cases (if you faced), or any meaningful insights

It must not exceed 10 pages, including tables and figures, excluding references. Do not make a cover page. It will be counted as a single page.

C. An open repository (e.g., Github) of your source code, and credits for contributions (means, who worked on which part must be clarified.)
D. Lightening talk presentation material for Dec 16th class, demo day. It should be within 3 minutes, which briefly introduces your approach and how it works. Then, you'll give additional 3 minutes for live (or video) demo.

2) Peer-review reports (Due on Dec 18) will be submitted individually via a link (will be) uploaded on LMS (after submission of final products).

4. Contacts for Q&A

For queries about deliverables, please contact the instructor (yongjaeyoo@hanyang.ac.kr) , and for queries regarding dataset, please contact our TA, Minho Chung (chungminho@hanyang.ac.kr).